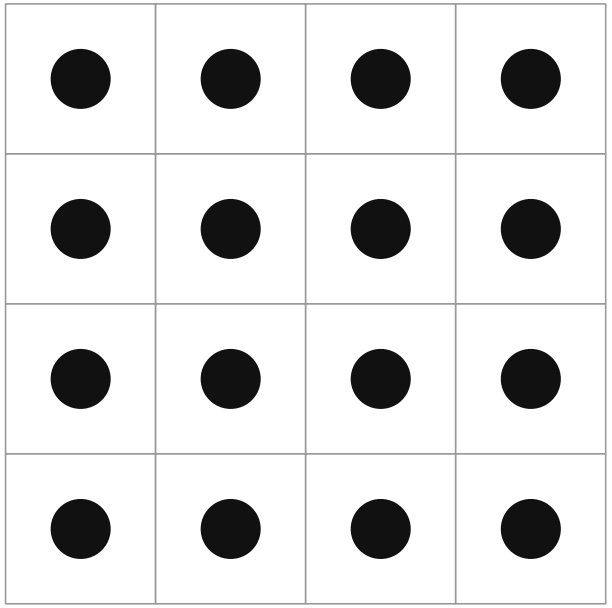
Expected Free Energy as Belief-Dependent Utility for ρ-POMDPs

Where is the target?
When should we stop looking?



4 × 4 Tileworld

One hidden target.
All cells equally likely.

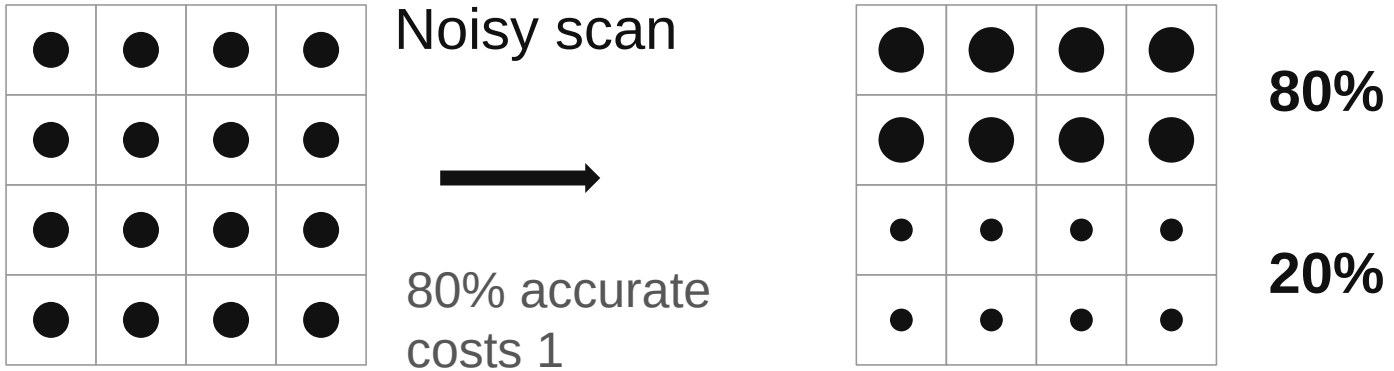
Patrick Cooper and Alvaro Velasquez

University of Colorado Boulder

EFE as belief-dependent utility

expected reward + w × information gain

Per-step utility in the specified reward-based recursion



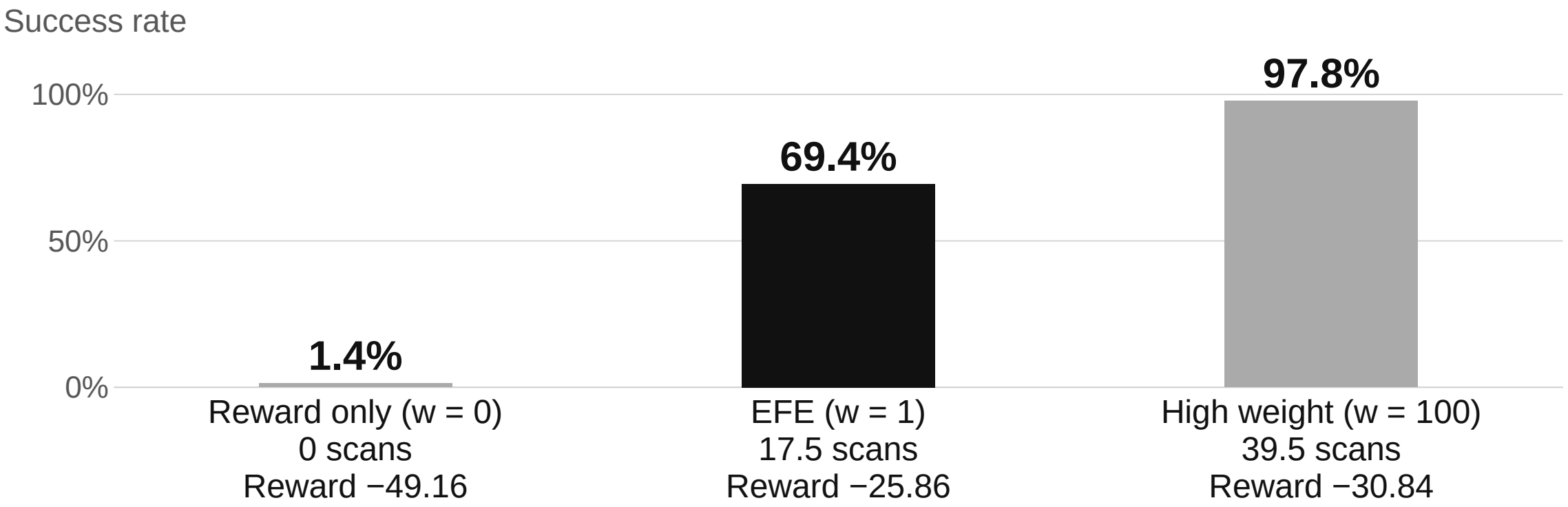
EFE uses w = 1

under the stated assumptions

Illustrative update. Dot area shows cell probability.

More certainty can cost more

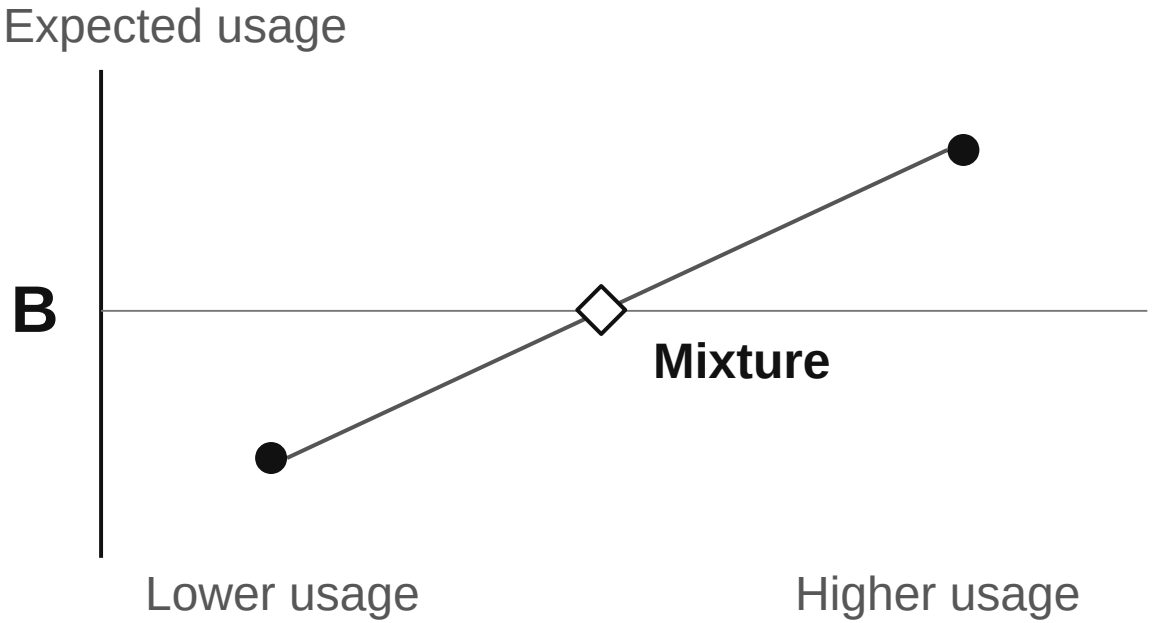
8 × 8 Tileworld. All three planners look ahead two steps.



Means over 5 seeds × 200 episodes. Reward includes scan costs.
At this horizon, reward-only planning takes no scans. EFE uses 1 reward unit per bit.

A target for expected sensing usage

Two policies can bracket the target



Schematic calibration

≤ 0.13

observations from target

Largest held-out mean error across 12 attainable cases. Tiger, Diagnosis and Bandit.

Calibration-derived targets. Per-episode usage can exceed B. Reward need not be optimal.